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بسم الله الرحمن الرحيم

Anatomy of the Birth canal & fetal skull

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- An understanding of the physiological and anatomical principles involved in normal and abnormal labour is best summarized using the **'3 Ps'** which are: the powers, the passages and the passenger.

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1. The 'powers' refers to forces, firstly the contractions of the uterine muscle that result in passage of the fetus through the birth canal, and secondly the maternal effort of pushing in the second stage of labour.

2. The 'passages' refers to the birth canal itself, which is made up of the bony pelvis, the muscles of the pelvic floor and the soft tissues of the perineum.

3. The 'passenger' refers to the fetus (size, presentation, lie, position).

- When the '3Ps' are favourable, normal labour is likely to ensue, resulting in an unassisted or spontaneous vaginal birth.

- When any of the '3Ps' are unfavorable, labour is likely to be abnormal, resulting in the need for intervention and with that, an increased risk of morbidity or mortality.

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The powers (uterine muscles):

- Myometrial cells has the ability to generate spontaneous contraction without nerve stimulation.

-It contain filaments of actin and myosin, which are the two key proteins for contraction.

-The interaction of myosin and actin brings about contraction, while their separation brings about relaxation, under the influence of intracellular free calcium.

-An increase in intracellular free calcium brings about contraction.

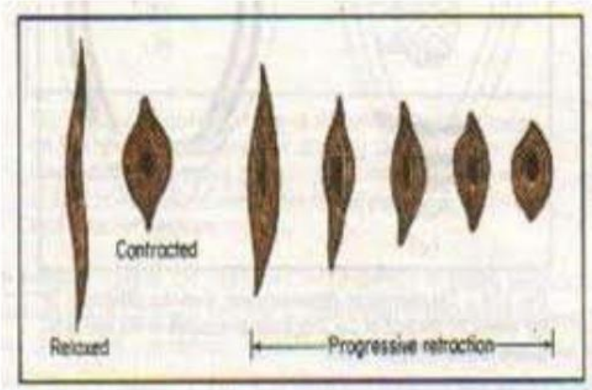
- Prostaglandins and oxytocin increase intracellular free calcium (so they stimulate uterine contraction).

- Beta- adrenergic compounds and calcium channel blockers decrease intracellular calcium (so they cause uterine muscle relaxation).

Contraction and retraction of myometrial cells

-Unlike any other muscle cell of the body, the uterine smooth muscle cells after contraction they relax but they do not return to their original length but become progressively shorter.

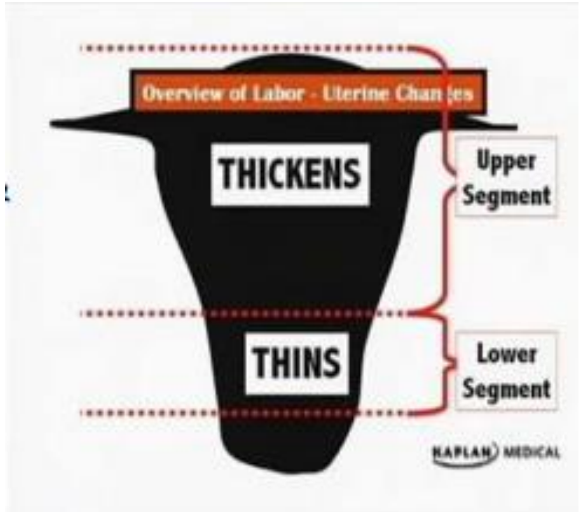
Contraction and retraction



-This unique feature of progressive shortening of the uterine smooth muscle cells is called **retraction** and occurs in the cells of the upper part of the uterus (body of uterus).

-The result of this retraction process is the development of the thicker, actively contracting body of uterus(upper segment).

-At the same time, the lower part of the uterus (isthmus and cervix) becomes thinner, dilated and stretched (called the lower segment).



What are the advantages of retraction of myometrial (uterine) muscle fibers?

1. During the 1st &2nd stage of labour, with each contraction and retraction the uterine cavity becomes progressively smaller and this will produce a downward pushing force on the fetus.

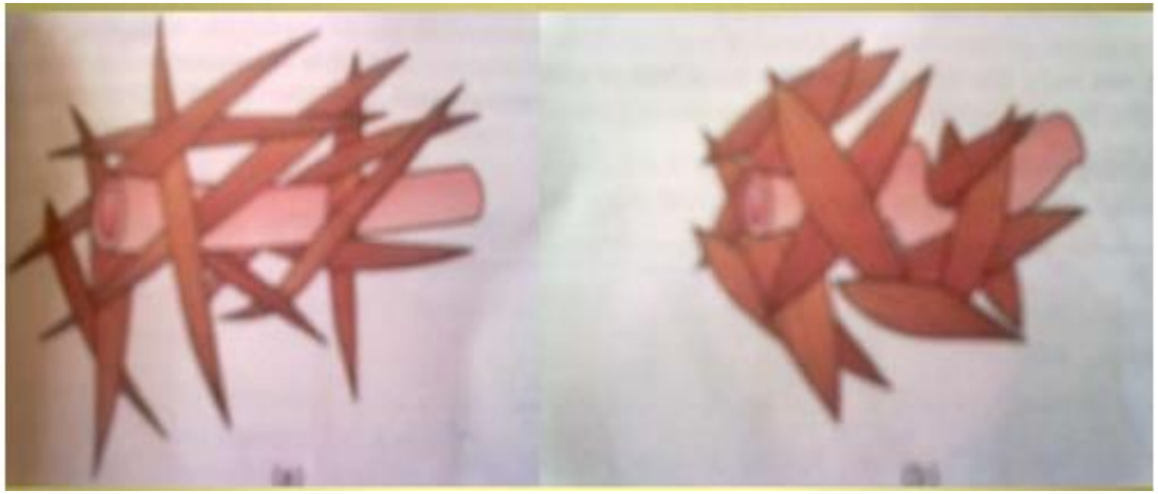
   -If contraction was followed by complete relaxation no progress and descend of the fetus will take place .

2. In the third stage of labour: after expulsion of the placenta,

contraction &retraction enables to close

the blood vessel sinuses at the placental bed

and prevent excessive blood loss (PPH).



The cervix:

- The Cervix is made up of collagen in a proteoglycan matrix with little smooth muscle cells.
- Throughout pregnancy the cervix is long, thick and closed, and this help to retain the fetus in the uterus up to the time of labour.
- Contractions at this point do not bring about effacement or dilatation.
- For birth to occur, and the cervix dilated easily in response to contractions, the cervix needs to be soft.
- This process of cervical softening called cervical Ripening and is an essential part of labour.
- Ripening usually happened before the onset of labour (1-2weeks).
- Ripening triggered by prostaglandins (PG E2 and PGF2α) and involve the following changes:

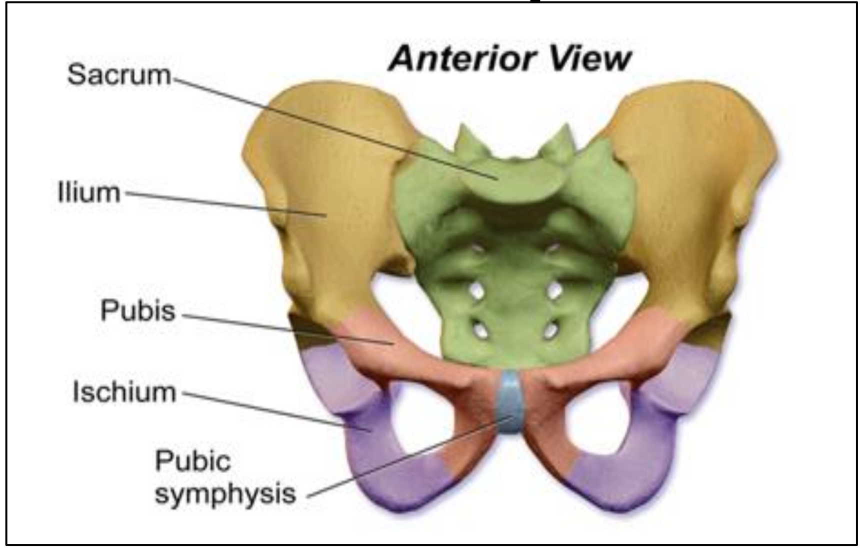
1. There is an increase in proteolytic activity and a reduction in collagen and elastin fibers.
2. Dermatan sulphate(which has high affinity to collagen) is replaced by the more hydrophilic hyaluronic acid, which results in, an increase in water content of the cervix.

All these changes make the cervical tissues soft and easily stretchable in response to uterine contractions which brings about effacement &dilatation.

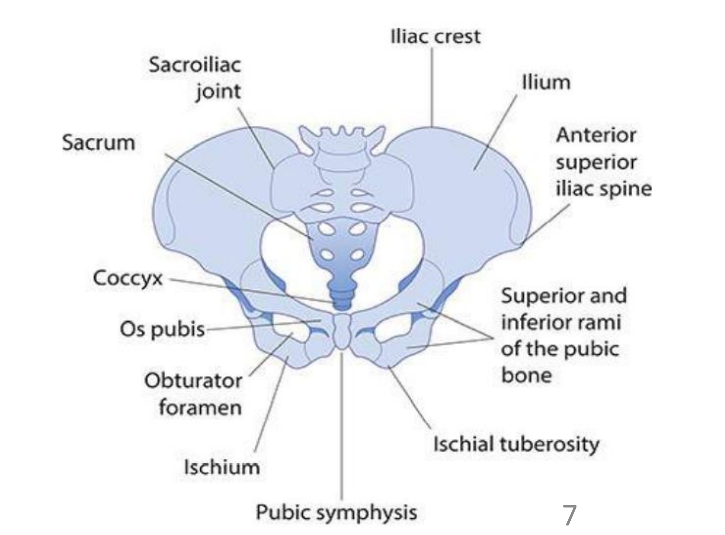
The 'passages':

The 'passages' refers to the birth canal itself, which is made up of the

1. Hard passages: the bony pelvis.
2. Soft passages composed of :
   - Cervix.
   - Muscles of the pelvic floor.
   - Soft tissues of the perineum.



- The pelvis is divided into:
- Greater (false) pelvis: Above pelvic brim; contains abdominal organs.
- Lesser (true) pelvis: Below pelvic brim; contains pelvic organs.
- The pelvis is made up of 4 bones:
- Two Hip bones (each consist of ilium, ischium and pubis.)
- Sacrum & Coccyx.



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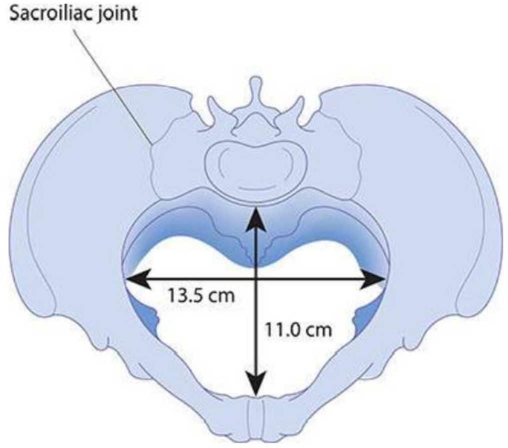
The true pelvis is divided into:

- The Inlet (pelvic brim).
- The mid-pelvis (the pelvic cavity).
- The Outlet.

• Each part has its diameters (transverse and antero-posterior) which affect the fetal descent through the birth canal during labour.

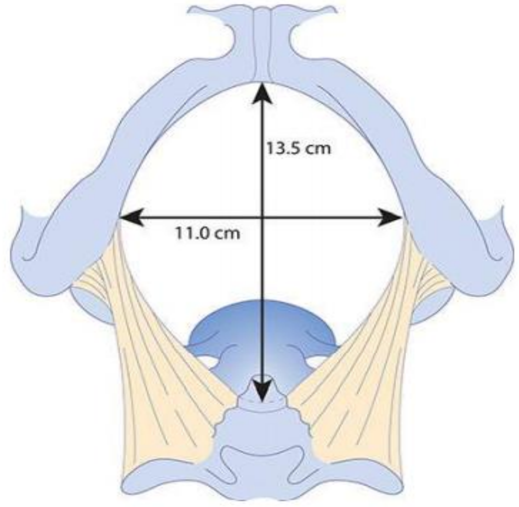
- **The pelvic inlet (brim):** characterized by
  - A wide transverse diameter = 13.5 cm.
  - Shorter anterior–posterior (A–P) diameter = 11.0 cm.



The pelvic brim.

- **The mid-pelvis (the mid-cavity):**
  - Is almost round.
  - The transverse and A-P diameters are similar at 12 cm.

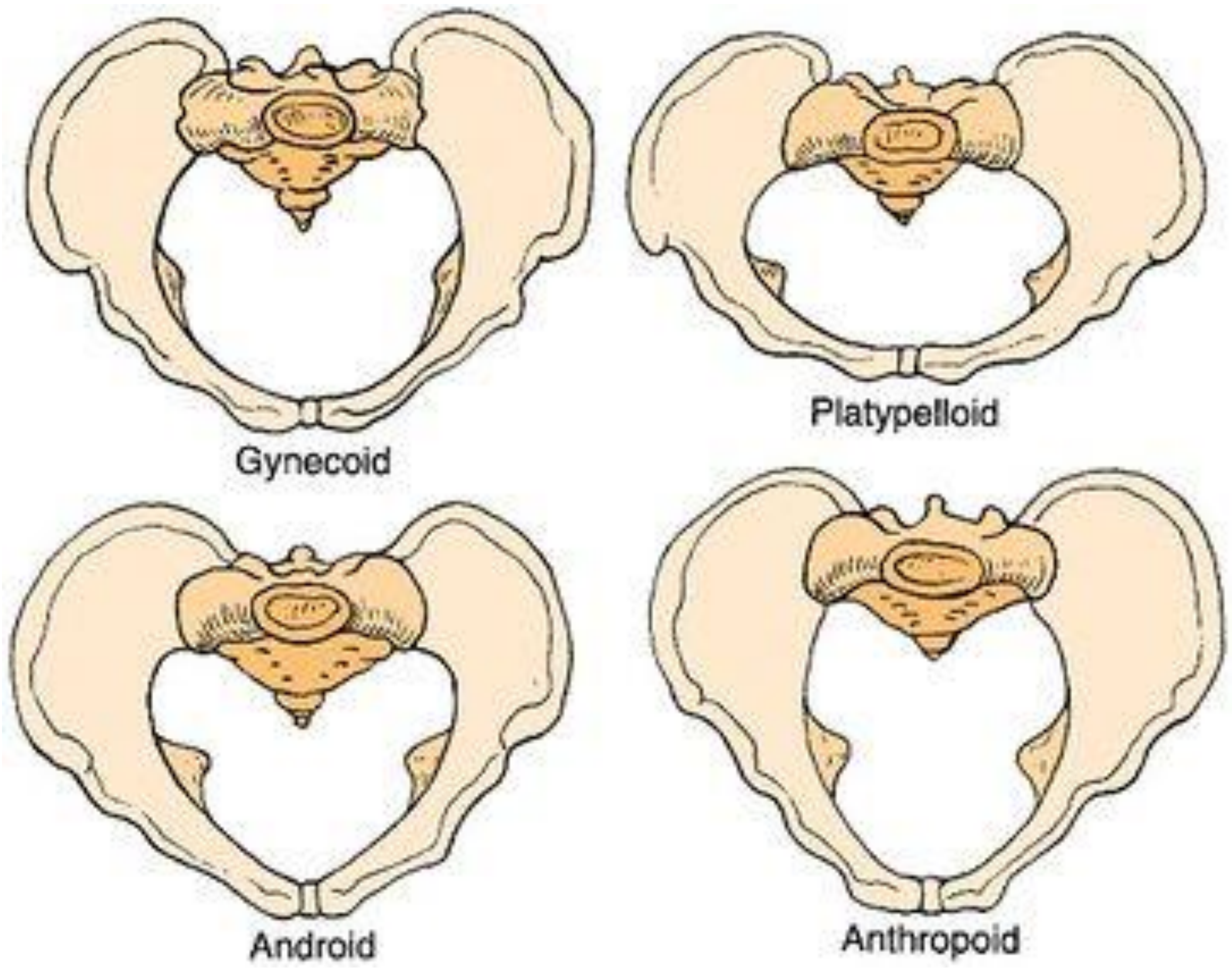
- **The pelvic outlet:**
  - Long A-P diameter = 13.5 cm.
  - Short transverse diameter =11 cm.



The pelvic outlet.

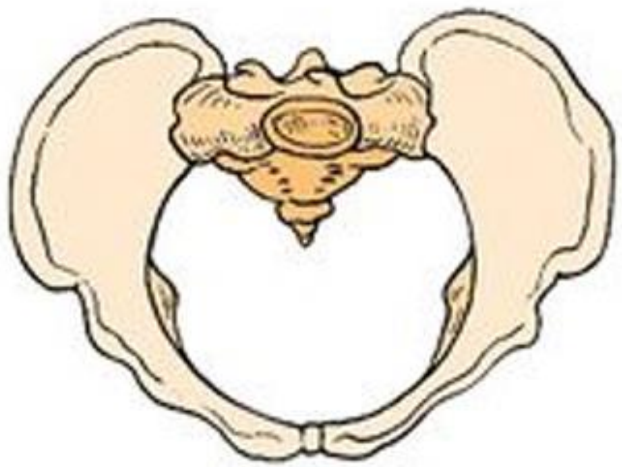
Pelvic Shape:

According to the shape, there are 4main pelvic types(Caldwell-Moloy)

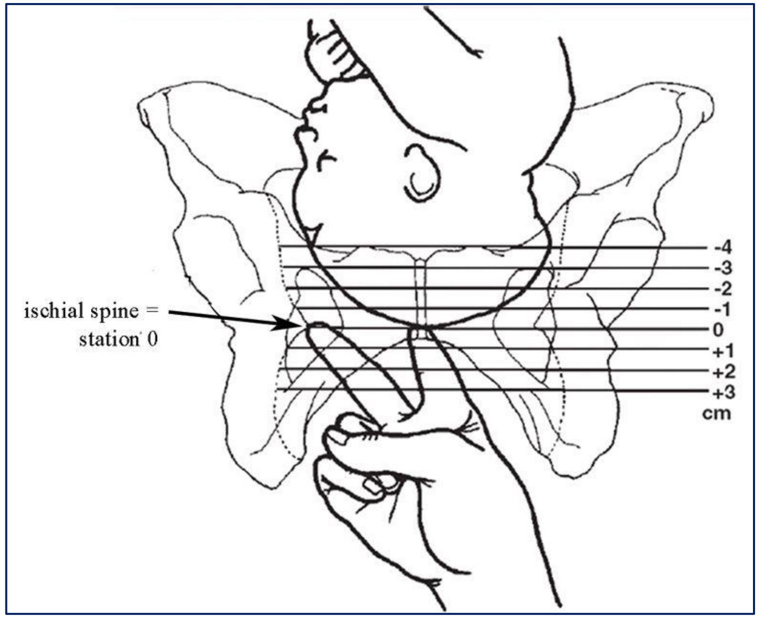


The gynecoid pelvis:

- The most common type(50%).
- It has:
- Round pelvic inlet.
- Shallow pelvic cavity.
- Short ischial spines.
- Curved sacrum.
- Wide subpubic angel.
- All these features allow easy descent and delivery of the fetus.
- So Gynecoid Pelvis is the most suitable pelvic shape for vaginal delivery.



- The ischial spines:
- Are two projections on lateral walls, palpable vaginally and are used as important landmarks for two purposes:



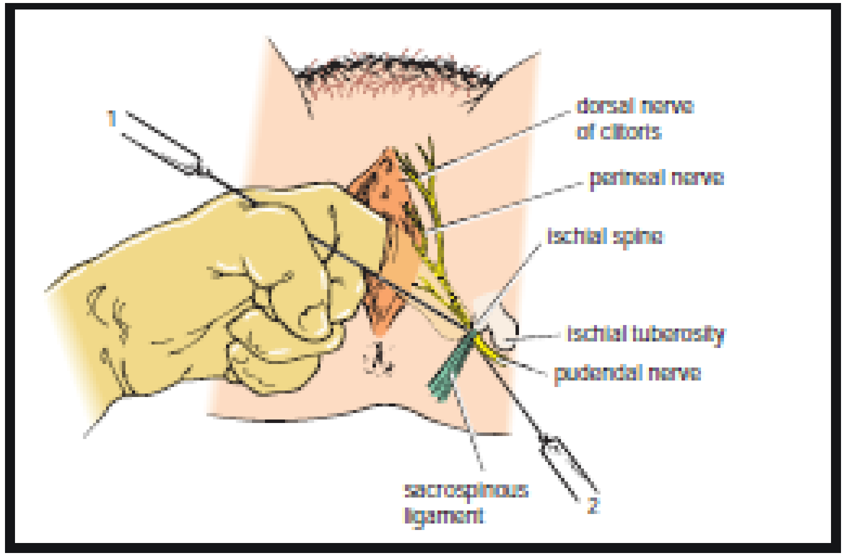
1. To assess the descent of the presenting part on vaginal examination:

*Station zero when the lower most part of the fetal head is at the level of the ischial spines

*–1 station when the lower most part of the fetal head is 1 cm above the spines and

*+1 station is 1 cm below the spine.

-Station zero is an important landmark clinically, it indicate engagement of the fetal head into the pelvis.



2. To provide a local anesthetic pudendal nerve block. The pudendal nerve passes behind and below the ischial spine on each side.

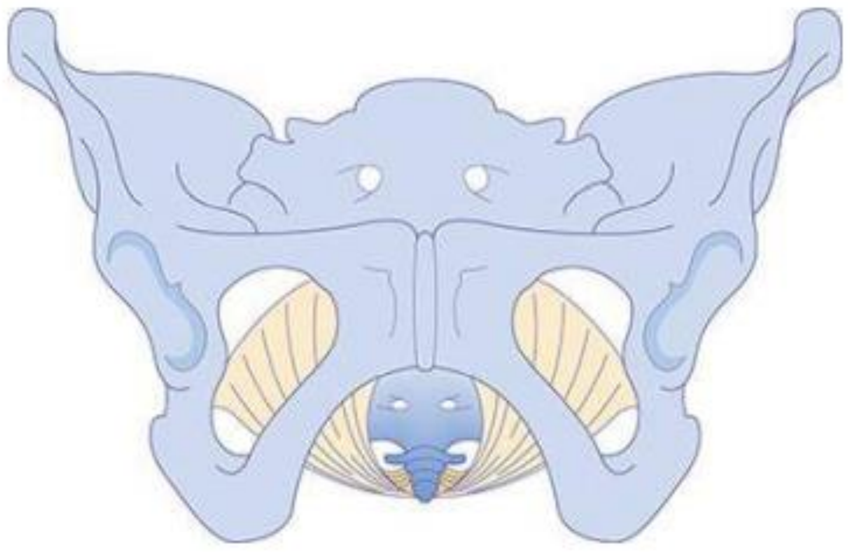
-A pudendal nerve block may be used when we assist delivery by instruments (e.g.: forceps-assisted delivery).

The pelvic floor

- This is formed by the two levator ani muscles which, with their fascia, form a musculo-fascial gutter during the second stage of labour.

- The configuration of the bony pelvis together with the gutter-shaped pelvic floor muscles:

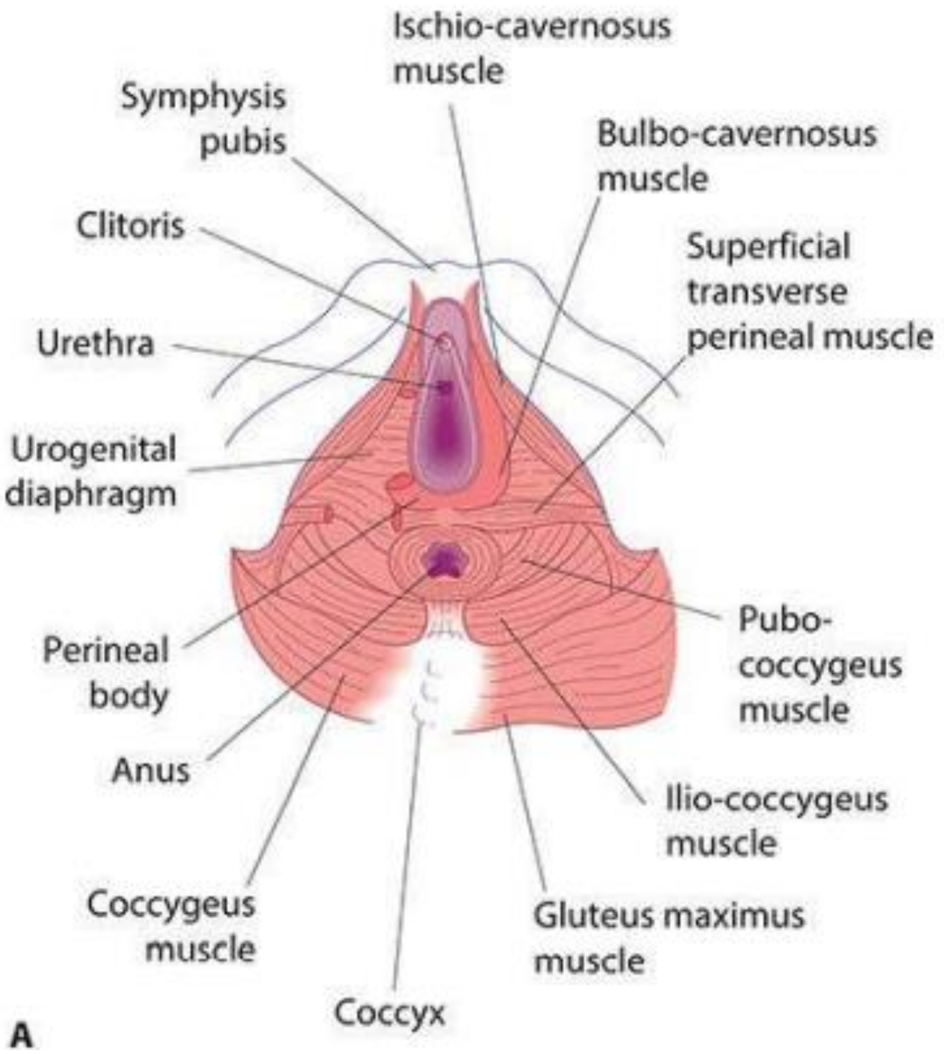
- Encourage the fetal head to flex and rotate as it descends through the mid-pelvis towards the pelvic outlet.



The musculofascial gutter of the levator sling.

The perineum

- The final obstacle to be overcome by the fetus during labour is the perineum.
- The perineal body is a condensation of fibrous and muscular tissue lying between the vagina and the anus.
- The perineum is taut and relatively resistant in the nulliparous woman, and pushing can be prolonged.
- Vaginal birth may result in tearing of the perineum and pelvic floor muscles or an episiotomy (surgical cut) may be required.
- In multiparous women, the perineum is stretchy and less resistant, resulting in faster labour and a higher probability of delivering with an intact perineum..



3.The passenger:

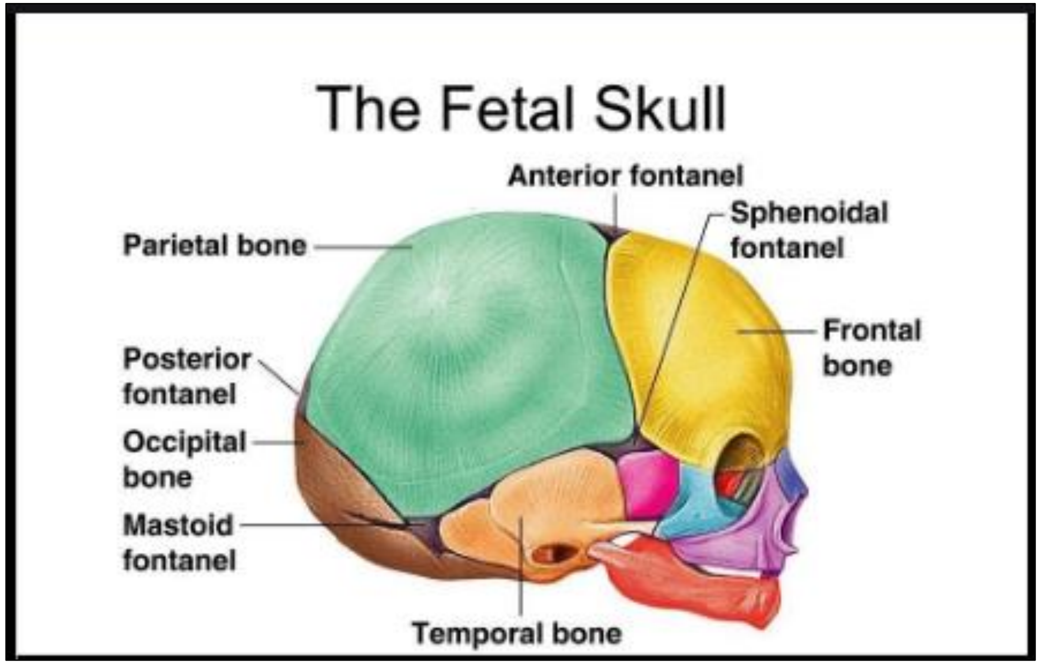
- **Refers to the fetus which affect the process of labour through its :**
- **Size (small, average, large).**
- **Presentation.**
- **Lie.**
- **Position.**
- **Attitude.**

The fetal skull

- **The fetal skull is made up of 3main parts: the vault, the face and the base.**

- **The sutures are the lines formed where the bony plates of the skull meet one another.**

- **Before delivery, the sutures joining the bones of the vault are soft, unossified membranes.**
- **Whereas the sutures joining the bones of the fetal face and the skull base are firmly united.**



The vault of the skull is composed of:

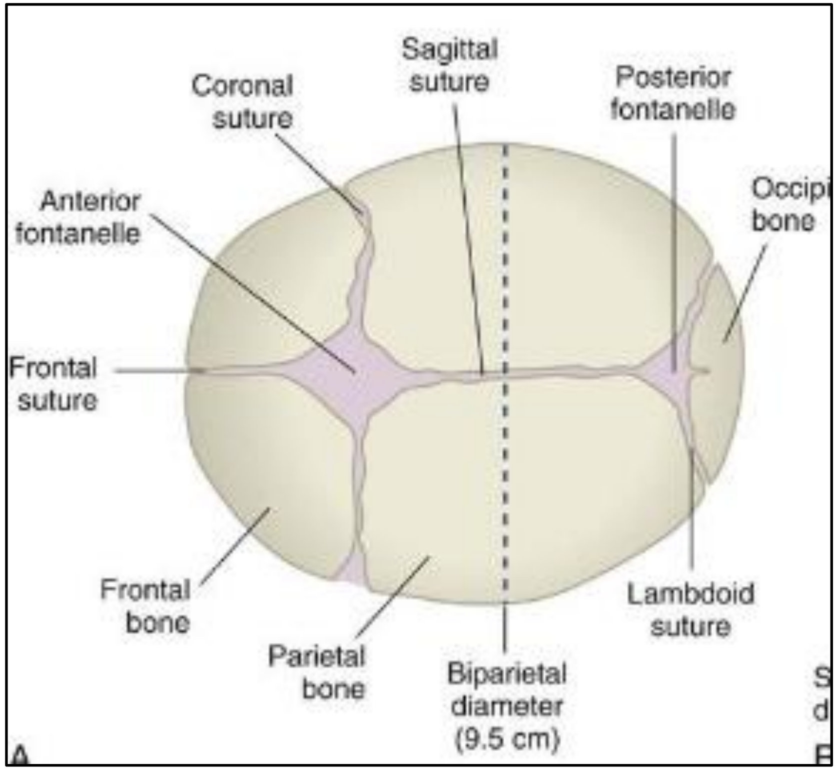
- **The two <u>parietal bones</u> and <u>parts</u> of the <u>occipital</u>, <u>frontal</u> and <u>temporal</u> bones.**
- **Between these bones there are <u>four membranous sutures</u>:**

*** <u>The sagittal suture</u>: between the 2 parietal bones.**

*** <u>The frontal suture</u>: between the 2parts of frontal bone.**

*** <u>The coronal suture</u>: between parietal & frontal bones.**

*** <u>The lambdoid suture</u>: between parietal & occipital bone.**



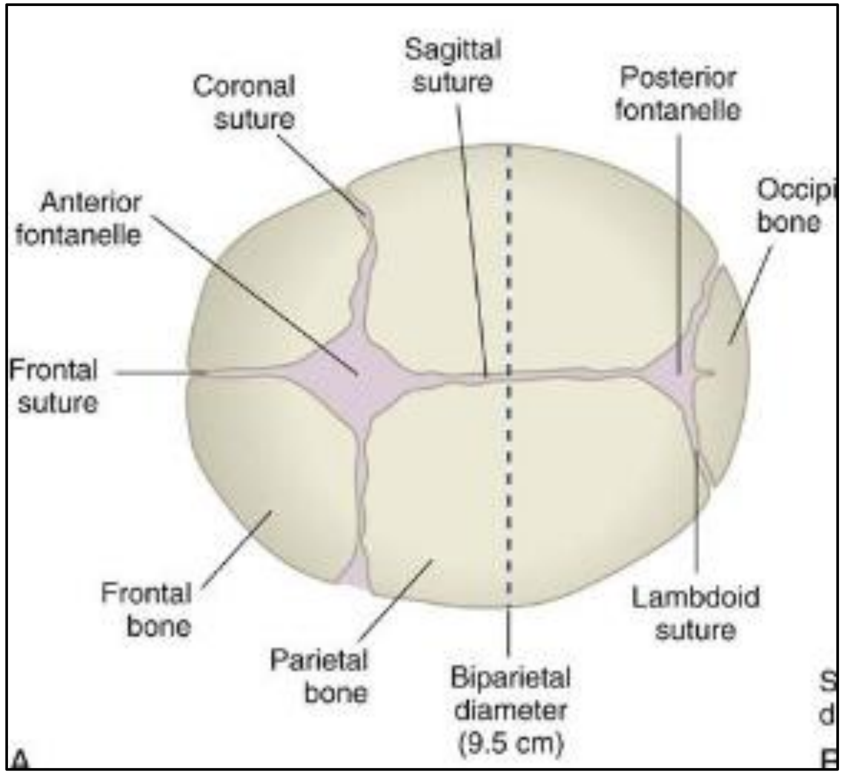
The membranous area that formed at the junctions of the various sutures is called fontanelle.

1. The anterior fontanelle (bregma):

- Is at the junction of the sagittal, frontal and two coronal sutures.

-It is diamond shaped.

- On vaginal examination four suture lines can be felt.



2. The posterior fontanelle:

- lies at the junction of the sagittal suture and the 2 lambdoid sutures between the two parietal bones and the occipital bone.

- It is smaller and triangular shaped.
- On vaginal examination three suture lines can be felt.

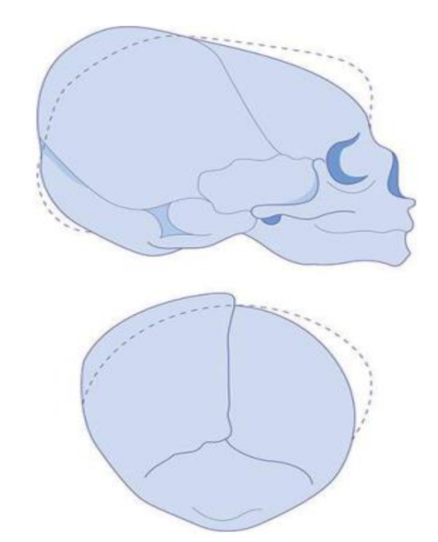
-The fact that the sutures are not fixed is important for labour.

-It allows the bones to move together and even to overlap.

-The parietal bones usually slide over the frontal and occipital bones.

This process of overlapping of the bones of the vault of the fetal skull during labour is called*Moulding*

-Moulding helps to reduce the diameters of the fetal head and encourages progress through the bony pelvis, while still protecting the underlying brain.



Schematic representation of moulding of the fetal skull.

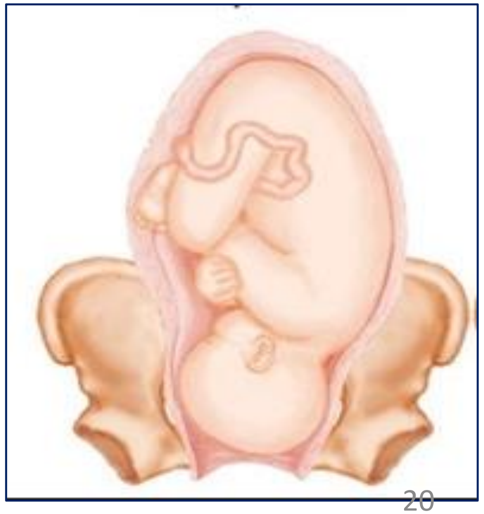
-However, severe moulding, or moulding early in labour, can be a sign of obstructed labour (means there is mismatch between the size of the fetal head and maternal pelvis called cephalo-pelvic disproportion.

Fetal presentation, lie, attitude & position:

-Presentation : refers to the part of the fetus occupying the lower part of the uterus and entering the pelvis first during labour.

-In 96% of pregnancies at term the fetus present with the head →cephalic presentation.

- When the head is well flexed on the body the vertex of the head will be the presenting part (vertex presentation).
- The vertex is the diamond - shaped area of the skull between the two parietal eminences laterally and the anterior and the posterior fontanelle.



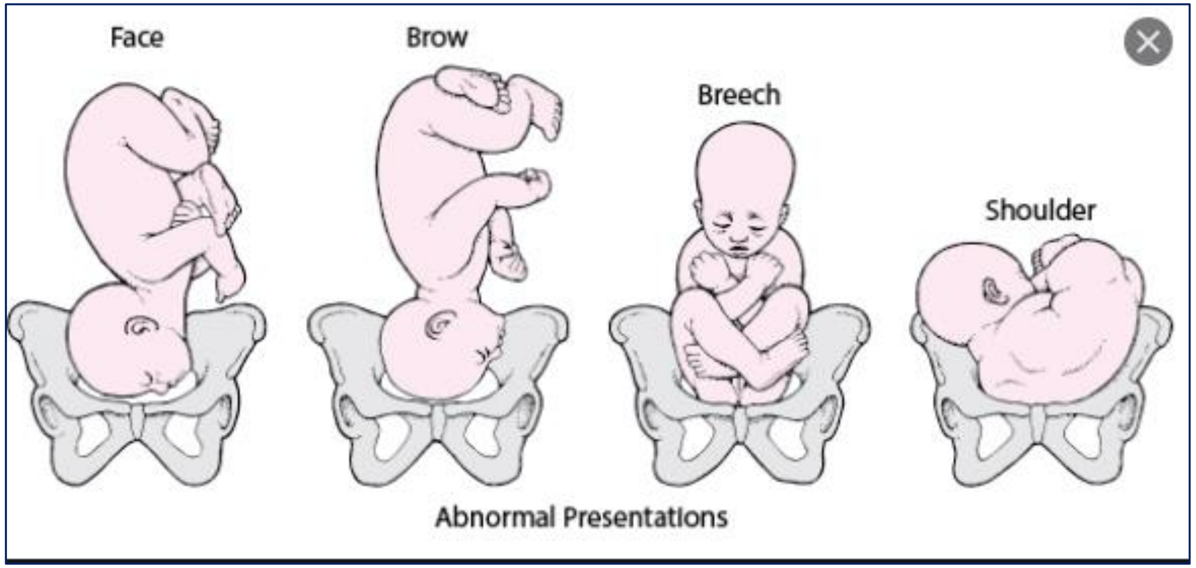
- So vertex presentation is the normal presentation which is compatible with normal vaginal delivery.
- Any presentation other than vertex is called malpresentation.
- **Types of malpresentation:**

- Breech presentation →the buttock of the fetus occupy the lower part of the uterus while the head occupying the fundus..

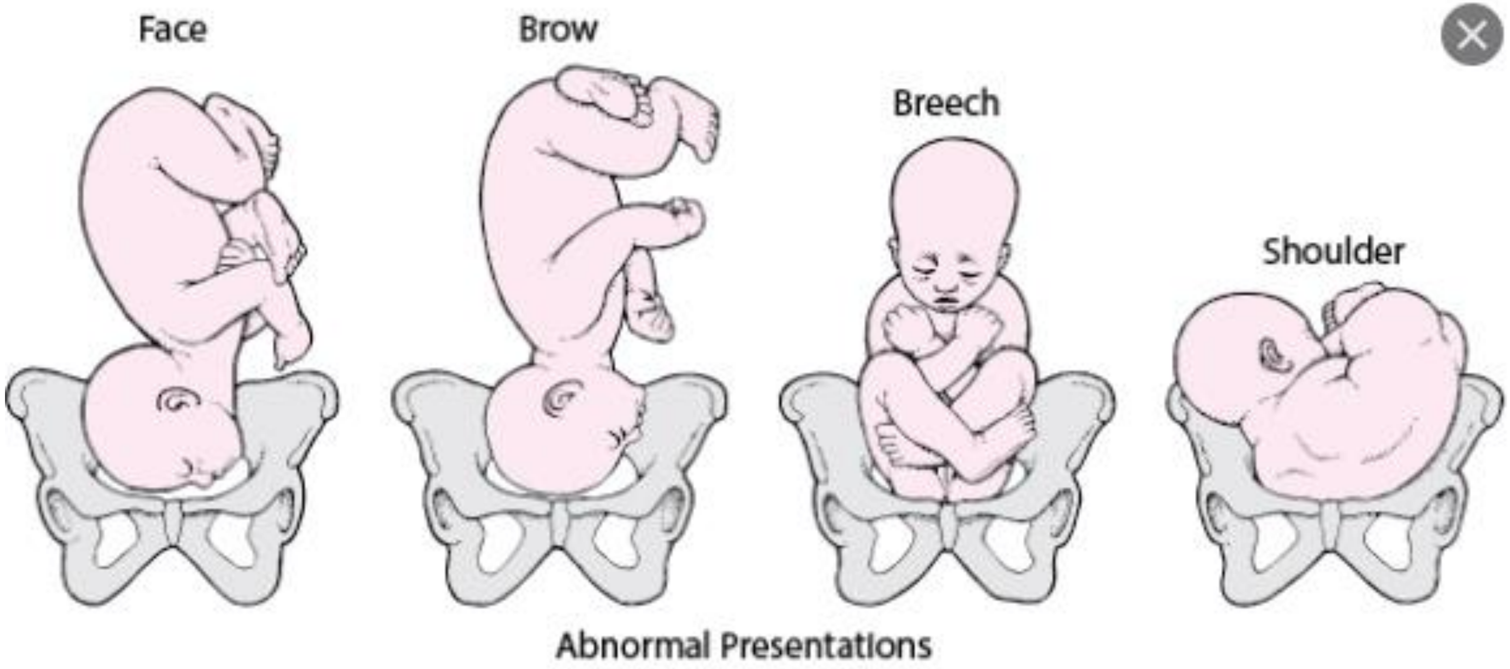
- Face presentation → it is cephalic but the head is completely extended on the body.

- Brow presentation → the head is partially extended and the forehead is the presenting part.

- Shoulder presentation → the fetus lying transversely and the shoulder of the fetus is the presenting part.



Vertex presentation and mal-presentations:



Lie of the fetus:

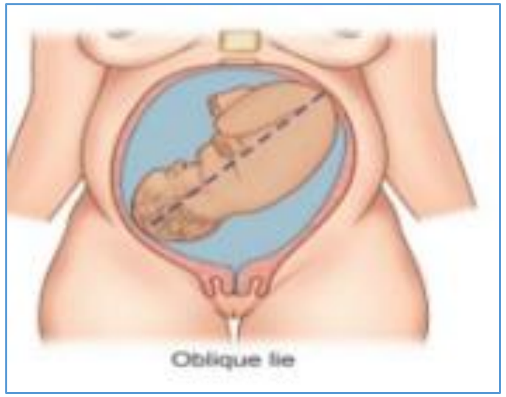
- it is Relation of long axis of the fetus to the long axis of the uterus:

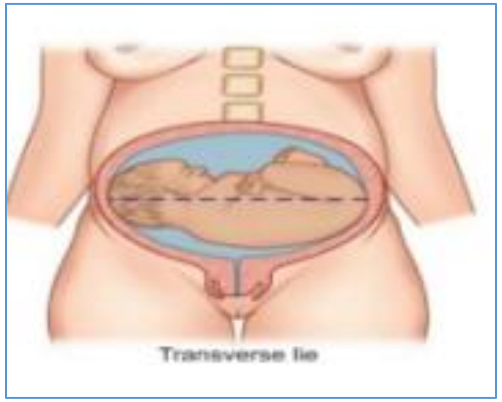
- Longitudinal (normal)

- Oblique (abnormal)

- Transverse(abnormal).

- Fetus with oblique or transverse lie can not be delivered vaginally.





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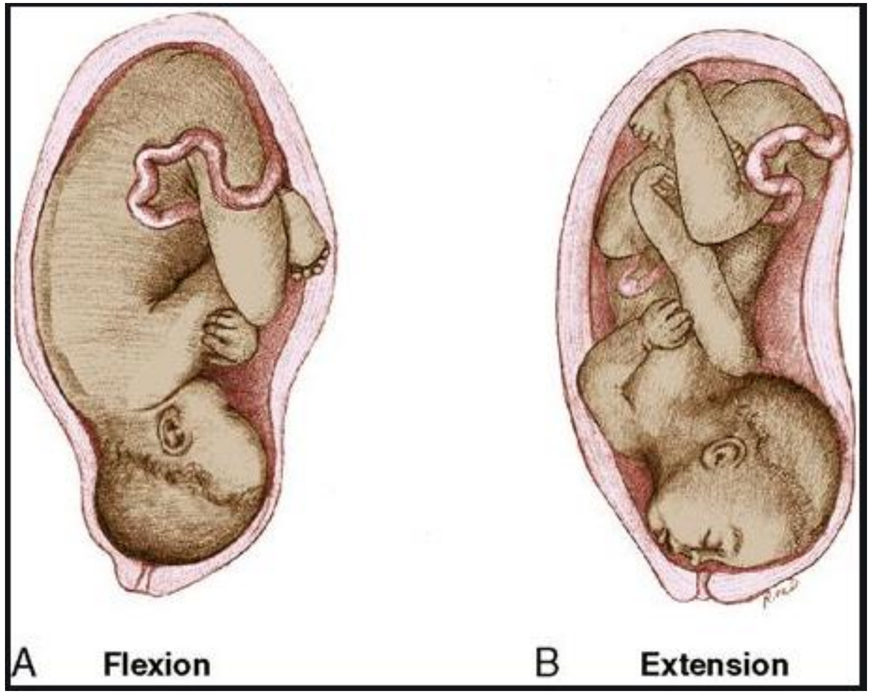
Attitude:

-Relation of the fetal parts to each other:

1.Flexion 2.Extension.

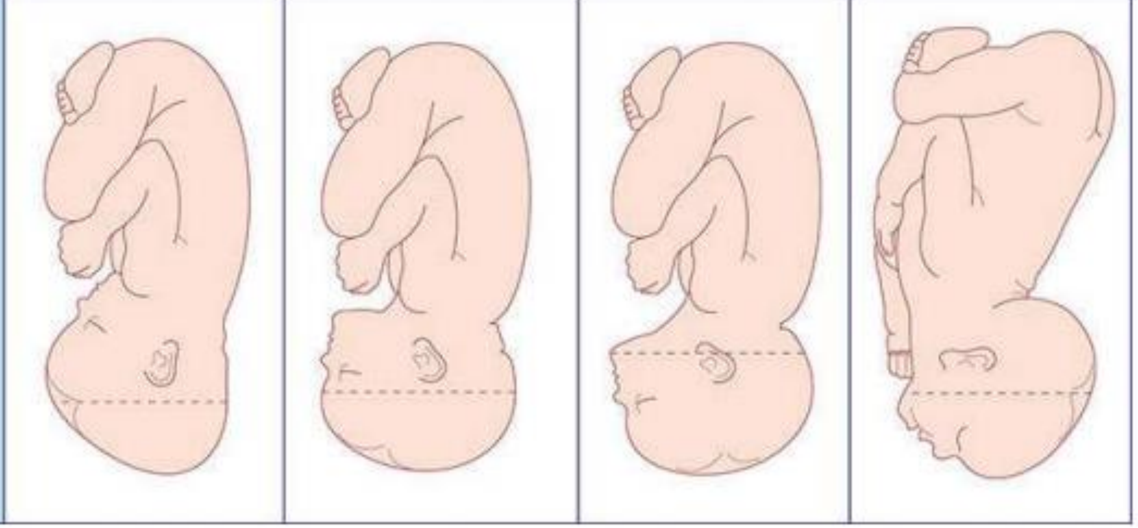
-Flexion brings smaller diameter of the fetal head to pass through the birth canal and help easy descent and delivery.



-The effect of fetal attitude on the presenting diameter.

- The more flexion of the head the smaller presenting diameter.

	Flexed → Extended			
Attitude	Well flexed	Less well flexed (partially extended) or deflexed	Extended 'brow presentation'	Hyperextended 'face presentation'
Diameter	Suboccipito-bregmatic	Occipito-frontal	Occipito-mental	Submento-bregmatic
Measurement	9.5 cm	11.5 cm	13.0 cm	9.5 cm

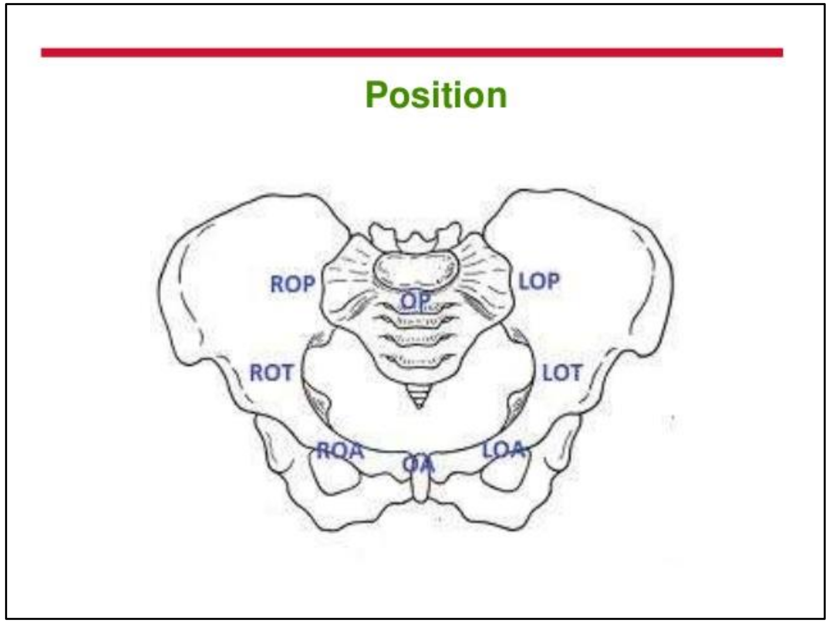


- **Position:** is defined as the orientation of the presenting part

in relation to the maternal symphysis pubis.

- Or it is the relationship of certain point on the presenting part (called denominator) to fixed points of the maternal pelvis.

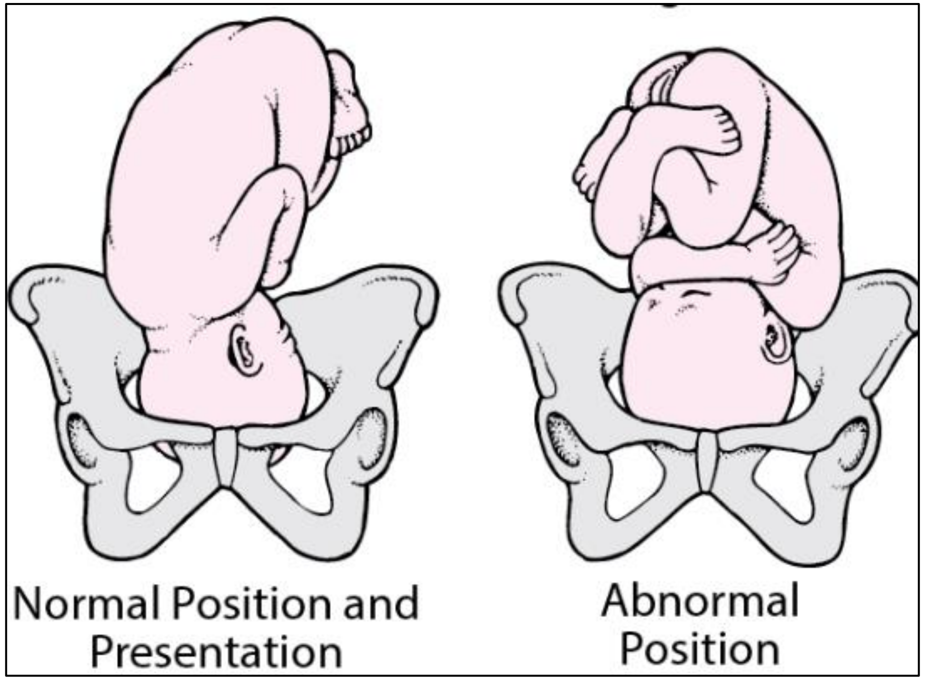
- The fixed points of the maternal pelvis are:
- The sacrum posteriorly.
- Sacro -iliac joint postero-laterally.
- Ileo-pectineal eminences antero-laterally.
- The symphysis pubis anteriorly.



- In vertex presentation The denominator is the occipital bone.

*Occipito-anterior position is normal.

*While occipito-posterior position is abnormal (malposition) and associated with prolonged labour.



In conclusion:

Failure to progress or abnormal labour can be due to abnormalities in the:

- **Powers:**

-Insufficient uterine contraction(can be corrected by giving oxytocin infusion)

- **Passages:**

– Abnormal bony pelvis.

– Rigid perineum.

- **Passenger (fetus) :**

– fetal size (Fetus too big).

– Abnormal fetal presentation (mal-presentation) e.g breech, brow, face.

- Abnormal fetal position (mal-position) e.g occiput posterior.
- Abnormal lie (oblique or transverse lie).


Thank you